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<Evaluating the Robustness of Lightweight Machine Learning Models to Label Noise>

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled **"Evaluating the Robustness of Lightweight Machine Learning Models to Label Noise"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Shikha Tuteja**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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CONTENT

Sr. No.	Topic	Page No.
1	Abstract	1
2	Introduction	1
3	Methodology	2
4	Tools & Technologies	4
5	Implementation	5
6	Major Findings/Outcomes/Output/Results	11
7	Conclusion & Future Scope	11
8	References	12

1. ABSTRACT

This study evaluates the robustness of six classical machine learning models — Logistic Regression, Decision Tree, Random Forest, SVM, KNN, and LightGBM — against label noise in training data. Experiments were conducted on two benchmark datasets, Breast Cancer Wisconsin and Adult Income, using symmetric and asymmetric noise injected at levels of 0%, 10%, 20%, and 30%. Performance was measured on clean test data using Accuracy and F1-score, along with relative degradation and fold-wise variance. Results show that symmetric noise causes greater performance degradation than asymmetric noise across all models and both datasets. SVM consistently demonstrated the strongest robustness under symmetric noise, while Decision Tree proved to be the most vulnerable model in nearly all settings. Notably, robustness rankings shifted under asymmetric noise, with KNN outperforming SVM on the Adult Income dataset, highlighting that robustness is context-dependent. The findings emphasize that clean-data accuracy alone is insufficient for model selection, and that robustness evaluation is essential when training label quality cannot be guaranteed.

2. INTRODUCTION

Statement of the Problem

In today's data-driven world, ensuring the quality of training labels in machine learning remains a persistent and often overlooked challenge. Many real-world datasets suffer from incorrect or corrupted labels due to human annotation errors, ambiguous instances, automated labeling pipelines, and inconsistencies in data collection processes. This label noise directly degrades model performance, leading to poor generalization and unreliable predictions. In critical domains such as healthcare, finance, and decision-support systems, such unreliable outputs can have serious consequences. Despite its significance, the impact of label noise on classical machine learning models is not always well understood, particularly in a controlled, comparative, and multi-dataset setting.

Background and Purpose of the Study

With the growing reliance on machine learning for tabular data classification, it is essential to understand how models behave when training labels are imperfect. Classical models such as Logistic Regression, Decision Trees, Random Forests, SVM, KNN, and LightGBM remain widely used in practice due to their simplicity and interpretability. However, their robustness under varying types and levels of label corruption has received limited systematic attention. This study addresses that gap by evaluating model behavior under two well-established noise settings — symmetric and asymmetric — across multiple noise levels and datasets, providing practical insights for model selection in noisy environments.

Overview of the Study

This research presents a controlled robustness evaluation framework applied to six classical machine learning models. Experiments are conducted on two benchmark datasets — the **Breast Cancer Wisconsin Diagnostic** dataset and the **Adult Income** dataset — representing contrasting data characteristics in terms of size, feature types, and class balance. Label noise is injected only into the training data at rates of **0%, 10%, 20%, and 30%**, while evaluation is always performed on clean test data using **Accuracy** and **F1-score** as primary metrics, along with relative degradation and fold-wise variance analysis. The study provides a systematic comparison that highlights which models remain stable, which degrade sharply, and how robustness depends on the interaction between model design, dataset characteristics, and the type of label corruption introduced.

3. METHODOLOGY

Research Objective

This study aims to systematically evaluate the robustness of six classical machine learning models under controlled label noise conditions on two benchmark tabular datasets. The core objective is to measure how model performance changes when incorrect labels are introduced into the training data, while the test data remains clean and untouched. The study compares degradation behavior across different models, datasets, noise types, and noise levels using standard classification metrics, providing a structured foundation for robustness-aware model selection.

Dataset Description

Two benchmark tabular classification datasets were selected for this study. The **Breast Cancer Wisconsin Diagnostic** dataset consists of 569 instances and 30 numerical features, representing a relatively small, clean, and well-structured binary classification problem sourced from the UCI Machine Learning Repository. The **Adult Income** dataset is significantly larger, containing 48,842 instances and 14 features of mixed types, including categorical and continuous variables, along with class imbalance and missing values. This dual-dataset setup ensures that findings are not specific to a single data distribution and improves the overall reliability and generalizability of the conclusions.

Model Selection

Six classical supervised machine learning models were selected to represent a diverse range of learning mechanisms. **Logistic Regression** serves as a linear baseline model. **Decision Tree** explores hierarchical rule-based splits and is known to be sensitive to training data variations. **Random Forest** applies ensemble averaging over multiple trees to improve stability. **Support Vector Machine (SVM)** constructs a maximum-margin hyperplane and is known for strong performance on structured tabular data. **K-Nearest Neighbors (KNN)** relies on instance-based voting and is inherently sensitive to noisy labels. **LightGBM** applies gradient

boosting through sequential tree construction and is considered one of the most powerful learners for tabular data.

Experimental Pipeline

The experimental workflow was designed as a fully controlled robustness evaluation pipeline. First, all six models were trained on the original clean data to establish a baseline reference performance. Next, **5-fold stratified cross-validation** was applied to ensure class proportions were preserved across all folds and to reduce dependence on any single train-test split. Label noise was then injected exclusively into the training labels at rates of **0%, 10%, 20%, and 30%**, while the test labels remained completely clean throughout all experiments. **Symmetric noise** was applied by randomly flipping a specified percentage of training labels uniformly across all classes, whereas **asymmetric noise** introduced directional label corruption from one class to another in a controlled manner. Each model was then trained on the noisy training set and evaluated on the clean test set for every combination of dataset, noise type, noise level, and fold.

Evaluation Metrics

Model performance was assessed using three complementary measures. **Accuracy** captures the overall proportion of correctly classified instances. **F1-score** provides a balanced measure of precision and recall, making it especially informative in the presence of class imbalance. **Relative Performance Degradation (RPD)** was computed as $RPD(\eta) = (M(0) - M(\eta)) / M(0) \times 100\%$, quantifying how strongly each model deteriorates as noise increases relative to its clean baseline. Additionally, the **standard deviation** of performance across cross-validation folds was analyzed to assess model stability under noisy supervision.

Implementation Details

All experiments were implemented in **Python** using a structured notebook-based pipeline. Separate notebooks were maintained for baseline evaluation, noise generation, cross-validation experiments, and result visualization for each dataset. A fixed random seed was applied throughout to ensure reproducibility of all results across runs.

4. Tools and Technologies

The following tools, technologies, and libraries were used throughout the design and implementation of this study:

Category	Tool / Technology	Purpose
Programming Language	Python 3.x	Core language used for all experiments, noise injection, model training, and result analysis
Development Environment	Jupyter Notebook	Notebook-based interactive pipeline maintained separately for baseline, noise, cross-validation, and plotting
Data Handling	Pandas	Dataset loading, cleaning, preprocessing, and structured manipulation of tabular data
Numerical Computing	NumPy	Array-level operations, random seed control, and mathematical computations across experiments
Machine Learning Framework	Scikit-learn	End-to-end implementation of Logistic Regression, Decision Tree, Random Forest, SVM, and KNN models
Gradient Boosting Library	LightGBM	Implementation and training of the LightGBM gradient boosting classifier for tabular data
Data Visualization	Matplotlib	Plotting accuracy and F1-score degradation curves, relative drop charts, and slope analysis figures
Statistical Visualization	Seaborn	Generating heatmaps, standard deviation bands, and robustness ranking visualizations
Model Evaluation	Scikit-learn Metrics	Computing Accuracy, F1-score, relative performance degradation, and fold-wise standard deviation
Cross-Validation	Scikit-learn StratifiedKFold	Applying 5-fold stratified cross-validation to preserve class proportions across all experimental folds

Noise Injection	Custom Python Functions	Designing and applying symmetric and asymmetric label noise at rates of 0%, 10%, 20%, and 30%
Reproducibility Control	NumPy and Scikit-learn Random Seed	Fixing random seed values to ensure fully consistent and reproducible results across all runs
Dataset Repository	UCI Machine Learning Repository	Source of both benchmark datasets — Breast Cancer Wisconsin Diagnostic and Adult Income
Version Control	GitHub	Managing experiment code, tracking changes, and maintaining organized project structure

This table covers all major tools used across data handling, model training, evaluation, and visualization in your research pipeline.

5. Implementation

This section presents the step-by-step implementation of the robustness evaluation framework, including the code, visual outputs, and result tables generated throughout the experimental pipeline.

5.1 Dataset Loading and Preprocessing

The Breast Cancer Wisconsin and Adult Income datasets were loaded and preprocessed before any experimentation. Preprocessing steps included handling missing values, encoding categorical variables, and normalizing feature values where required.

In [5]:
```
# cell 2

import numpy as np
import pandas as pd

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split

from models import get_models
from evaluation import evaluate_model

np.random.seed(42)
```

In [6]:
```
# cell 3

data = load_breast_cancer()
X = data.data
y = data.target

print("Dataset shape:", X.shape)
```

```
Dataset shape: (569, 30)
```

In [5]:
```
# cell 4

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("Train size:", X_train.shape)
print("Test size:", X_test.shape)
```

```
Train size: (455, 30)
Test size: (114, 30)
```

```python
data = load_breast_cancer()
y = data.target

print("Original class distribution:")
print("Class 0:", np.sum(y == 0))
print("Class 1:", np.sum(y == 1))
```

```
Original class distribution:
Class 0: 212
Class 1: 357
```

```python
y_sym = add_symmetric_noise(y, noise_rate=0.2)

print("After 20% Symmetric Noise:")
print("Class 0:", np.sum(y_sym == 0))
print("Class 1:", np.sum(y_sym == 1))
```

```
After 20% Symmetric Noise:
Class 0: 241
Class 1: 328
```

```python
y_asym = add_asymmetric_noise(y, noise_rate=0.2)

print("After 20% Asymmetric Noise (1 -> 0):")
print("Class 0:", np.sum(y_asym == 0))
print("Class 1:", np.sum(y_asym == 1))
```

```
After 20% Asymmetric Noise (1 -> 0):
Class 0: 283
Class 1: 286
```

Fig 5.1 : Code for Loding and Processing

5.2 Noise Injection

Custom Python functions were implemented to inject symmetric and asymmetric label noise into the training labels at rates of 0%, 10%, 20%, and 30%, while keeping the test labels completely clean.

```python
def add_symmetric_noise(y, noise_rate, random_state=42):


    np.random.seed(random_state)
    y_noisy = y.copy()

    n_samples = len(y)
    n_noisy = int(noise_rate * n_samples)

    noisy_indices = np.random.choice(n_samples, n_noisy, replace=False)

    for idx in noisy_indices:
        y_noisy[idx] = 1 - y_noisy[idx]  # binary flip

    return y_noisy
```

Fig 5.2 : Noise Injection Table

5.3 Model Training

All six classical machine learning models were trained on the noisy training data for each noise type and noise level combination using 5-fold stratified cross-validation.

	Model	Accuracy	F1
0	Logistic Regression	0.982456	0.986111
1	Decision Tree	0.912281	0.928571
2	Random Forest	0.956140	0.965517
3	SVM	0.982456	0.986111
4	KNN	0.956140	0.965517
5	LightGBM	0.964912	0.972603

Fig 5.3 : Model Training Table

5.4 Baseline Results

The table below presents the baseline accuracy and F1-score of all six models evaluated on clean data before any noise was introduced.

Cross-Dataset Comparison of Baseline Performance and Noise Robustness

Dataset	Best Baseline Model	Best Symmetric Robustness Model	Best Asymmetric Robustness Model	Worst Overall Model Under Noise	Strongest F1 Collapse
Breast Cancer	SVM	SVM	SVM	Decision Tree	Decision Tree
Adult Income	LightGBM	SVM	KNN	Decision Tree	Decision Tree / Logistic Regression / SVM

Fig 5.4 : Comparison Table

5.5 Symmetric Noise Results

The graphs and tables below illustrate the accuracy and F1-score degradation of all models under increasing levels of symmetric label noise on both datasets.

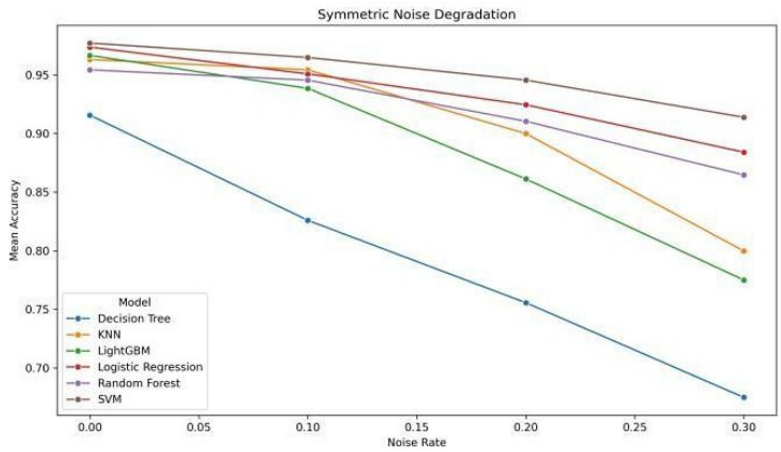

Fig 5.5. : Accuracy degradation under symmetric noise across noise levels — Breast Cancer Wisconsin dataset

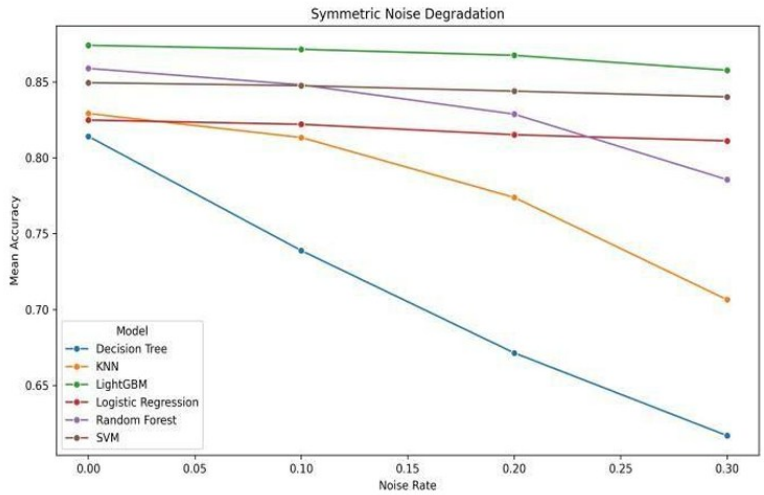

Fig 5.5.1 : Accuracy degradation under symmetric noise across noise levels — Adult Income dataset.

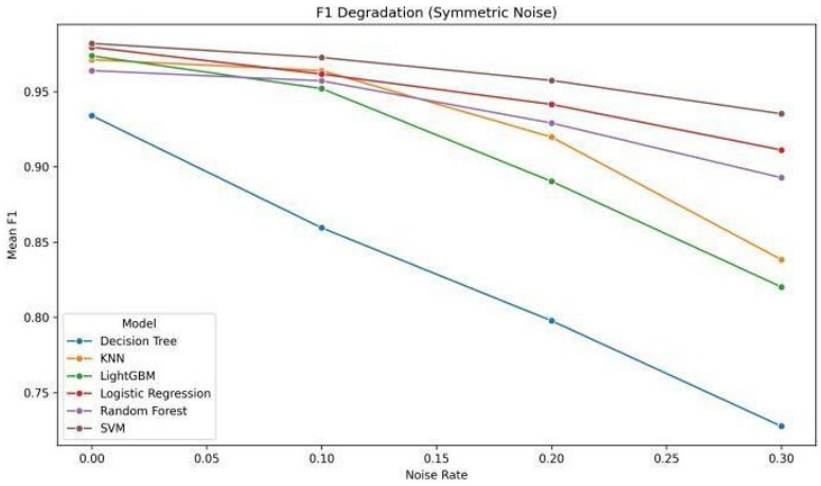

Fig. 5.5.2 : F1-score degradation under symmetric noise — Breast Cancer Wisconsin dataset.

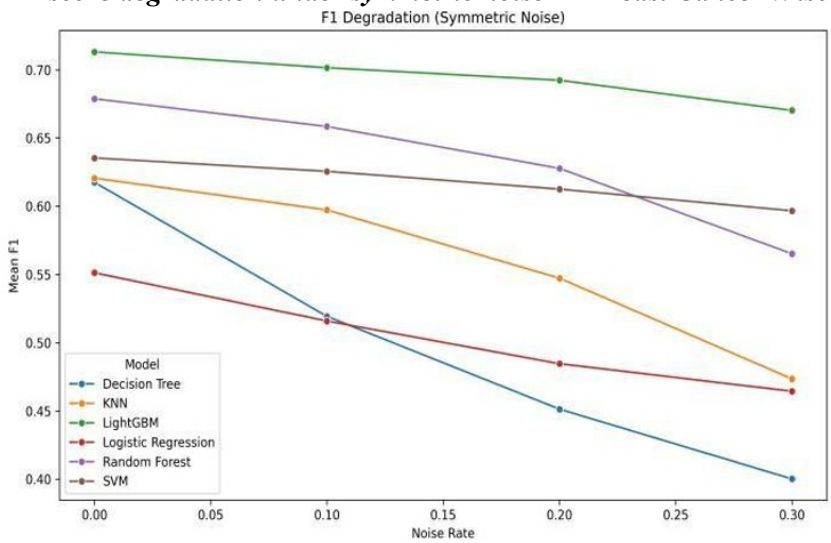

Fig 5.5.3 : F1-score degradation under symmetric noise — Adult Income dataset.

5.6 Asymmetric Noise Results

The graphs and tables below illustrate the accuracy and F1-score degradation of all models under increasing levels of asymmetric label noise on both datasets.

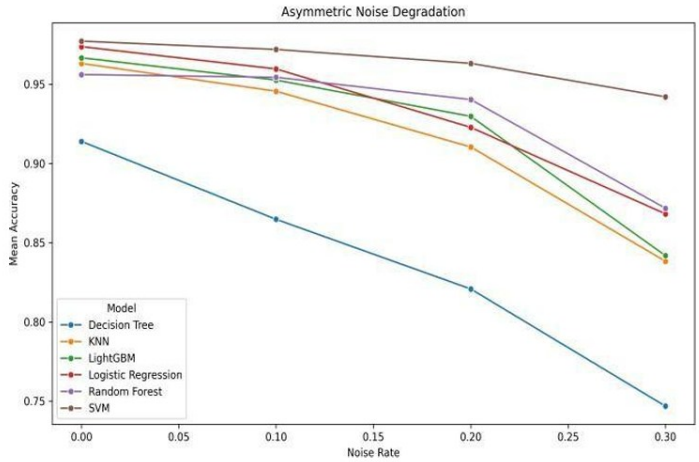

Fig 5.6 : Accuracy degradation under asymmetric noise across noise levels — Breast Cancer Wisconsin dataset.

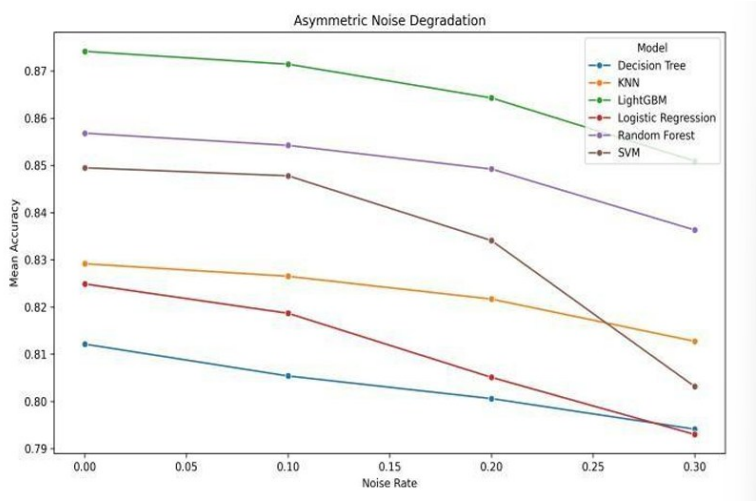

Fig 5.6.1 : Accuracy degradation under asymmetric noise across noise levels — Adult Income dataset.

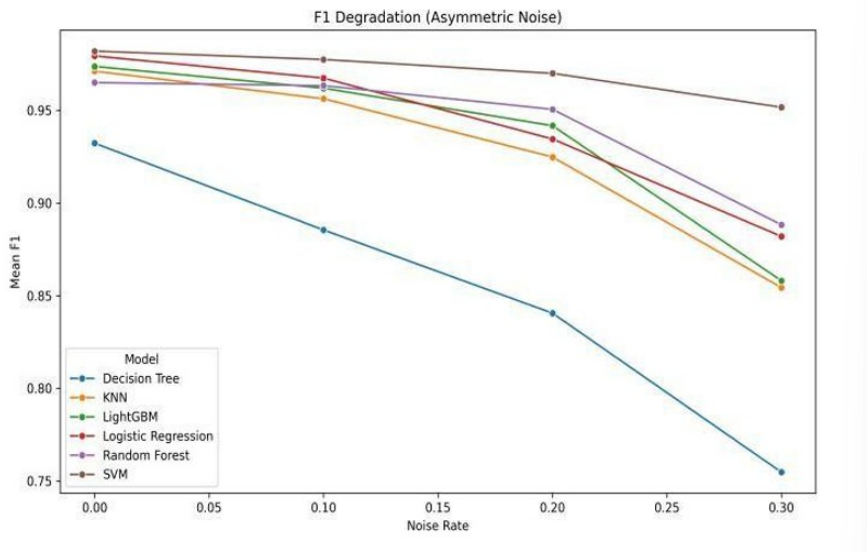

Fig 5.6.2 : F1-score degradation under asymmetric noise — Breast Cancer Wisconsin dataset.

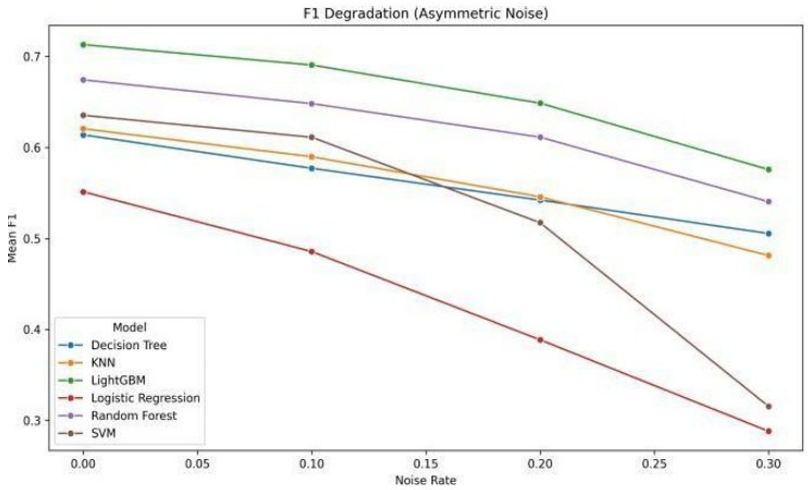

Fig 5.6.3 : F1-score degradation under asymmetric noise — Adult Income dataset.

5.7 Robustness Ranking and Comparative Analysis

The heatmaps, ranking charts, and degradation slope graphs below summarize the overall robustness of each model across both noise types and both datasets.

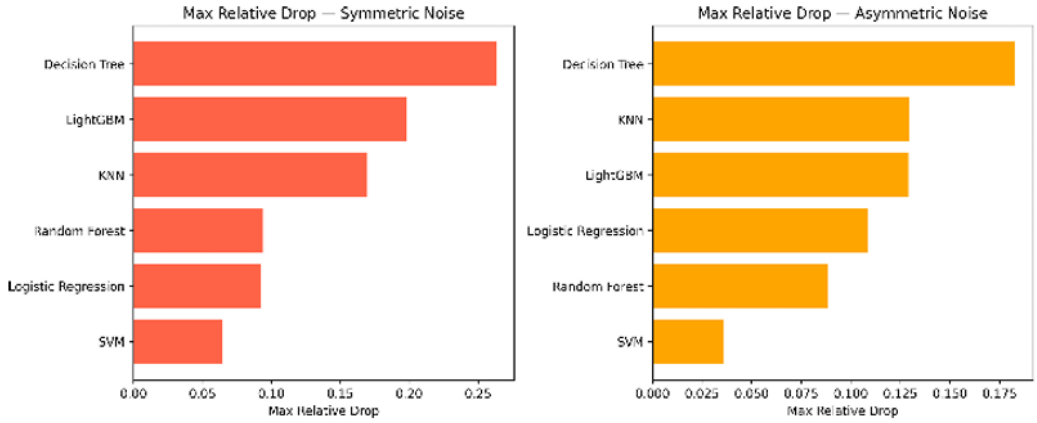

Fig 5.7 : Robustness ranking of all models by maximum relative accuracy drop at the highest noise level (30%) under symmetric (left) and asymmetric (right) noise - Breast Cancer Wisconsin.

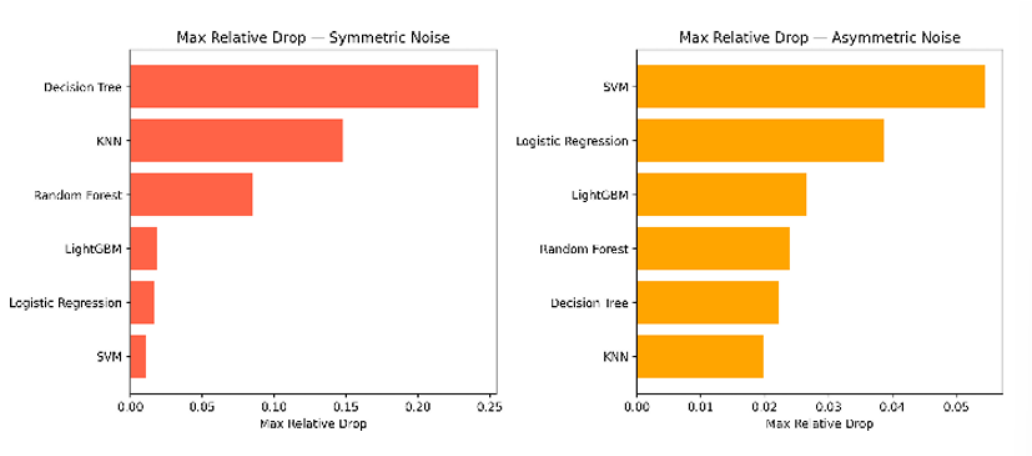

Fig 5.7.1 : Robustness ranking of all models by maximum relative accuracy drop at the highest noise level (30%) under symmetric (left) and asymmetric (right) noise – Adult Income Dataset

6. Major Findings / Outcomes / Output / Results

Criterion	Breast Cancer	Adult Income	Pattern
Best Baseline Accuracy	SVM (97.7%)	LightGBM (87.4%)	Dataset-dependent
Best Sym. Robustness	SVM (6.5% rel. drop)	SVM (1.1% rel. drop)	Consistent: SVM
Best Asym. Robustness	SVM (3.6% rel. drop)	KNN (2.0% rel. drop)	Noise-type dependent
Least Robustness	Decision Tree (26.3%)	Decision Tree (24.2%)	Consistent: DT worst
Strongest F1 Collapse	Decision Tree	Logistic Reg. / SVM	Asym. noise on Adult

Fig 6 : Combined cross-dataset comparison showing best baseline, best robustness under each noise type, worst model, and F1 collapse pattern.

The experiments conducted across both datasets revealed several important findings regarding the robustness of classical machine learning models under label noise.

- **Label noise degrades all models**, but the degree of degradation varies significantly depending on the model type, dataset characteristics, and the type of noise introduced.
- **Symmetric noise is more destructive** than asymmetric noise across all models and both datasets. Random label flipping in all directions eliminates more class structure, resulting in consistently higher relative performance drops.
- **SVM is the most robust model under symmetric noise**, achieving only a 6.5% relative accuracy drop on the Breast Cancer dataset and just 1.1% on the Adult Income dataset at 30% noise level.
- **Decision Tree is consistently the least robust model**, showing the highest relative accuracy drops exceeding 24% on both datasets under both noise types, along with high fold-wise variability.
- **Robustness is context-dependent and not universal.** Under asymmetric noise on the Adult Income dataset, KNN emerged as the most robust model with only a 2.0% relative drop, overtaking SVM which became the most affected model in that specific setting.
- **Clean-data accuracy does not imply robustness.** LightGBM achieved the best baseline accuracy on the Adult Income dataset but ranked lower than SVM in symmetric noise robustness.
- **F1-score reveals hidden degradation** more clearly than accuracy, particularly on the imbalanced Adult Income dataset, where some models showed moderate accuracy decline but a sharp collapse in F1-score under asymmetric noise.
- **Ensemble and margin-based models** such as Random Forest, LightGBM, and SVM demonstrated comparatively stable and consistent performance across most noise settings, while tree-based and instance-based models showed greater sensitivity.

7. Conclusion and Future Scope

Conclusion

This study systematically evaluated the robustness of six classical machine learning models — Logistic Regression, Decision Tree, Random Forest, SVM, KNN, and LightGBM — under controlled symmetric and asymmetric label noise on two benchmark tabular datasets. The findings clearly demonstrate that label noise consistently reduces model performance, and that the degree of degradation is strongly influenced by the learning mechanism, dataset characteristics, and the type of corruption introduced. SVM proved to be the most robust model under symmetric noise across both datasets, while Decision Tree was the most vulnerable in nearly all settings. The study also established that F1-score is a more sensitive and informative metric than accuracy alone in noisy-label evaluations, particularly under class imbalance. Overall, the results confirm that robustness evaluation is an essential component of model selection in any real-world scenario where training label quality cannot be fully guaranteed.

Future Scope

This work can be extended in several meaningful directions in the future. Additional benchmark datasets from diverse domains can be included to strengthen the generalizability of the findings. More realistic noise mechanisms such as class-dependent noise, instance-dependent noise, and real annotation uncertainty can be studied to better reflect practical data collection scenarios. The evaluation framework can be expanded to include deep learning models and noise-aware training strategies such as loss correction, sample reweighting, confident learning, and noise filtering. Furthermore, statistical significance testing and theoretical analysis can be incorporated to provide deeper interpretability of why certain models remain more stable than others under

specific corruption settings. This study serves as a solid experimental foundation for more advanced and application-oriented research in noisy supervised learning.

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